

State-Level Renewable Energy Production and the 2022 Energy Price Shock: Heterogeneous Effects on U.S. Household Energy Expenditure

A Difference-in-Differences Approach Using the Consumer Expenditure
Survey

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Abstract

This paper investigates whether the 2022 energy price shock—triggered by the Russian invasion of Ukraine and subsequent disruptions to global fossil fuel markets—had a differential impact on household energy expenditure across U.S. states with varying levels of renewable energy production per capita. Using household-level microdata from the U.S. Consumer Expenditure Survey (CES) spanning 2019Q1 to 2024Q1, we implement a Difference-in-Differences (DiD) framework with continuous treatment intensity. Our full-sample results show no statistically significant average treatment effect ($\beta = -0.006$, $p = 0.39$), suggesting that, on average, state-level renewable energy production did not measurably buffer households from the price shock. However, heterogeneity analysis reveals two suggestive findings that warrant further investigation. First, high-income households (top quartile) in states with greater renewable energy production experienced a reduction in energy expenditure of approximately 4.5 percent post-2022 ($\beta = -0.045$, $p = 0.004$), with pre-trends validated by a Wald test ($p = 0.20$). Second, higher renewable production intensity is associated with a modest reduction in the probability of annualized energy burden—defined as annual energy expenditure exceeding 10 percent of household income—of 0.35 percentage points per standard deviation ($\beta = -0.004$, $p = 0.018$). These heterogeneous effects are consistent with differences in housing tenure, market exposure, and the availability of energy assistance programs across the income distribution. We discuss the limitations of our treatment proxy and the absence of weather controls, which qualify the strength of causal interpretation.

Keywords: Energy expenditure, renewable energy, difference-in-differences, energy price shock, Consumer Expenditure Survey, household heterogeneity, energy burden

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1 Introduction

The sharp increase in energy prices observed in 2022 represented one of the most significant global economic shocks in recent years, driven by a combination of post-pandemic demand recovery, supply chain disruptions, and geopolitical tensions. In the United States, this shock translated into substantial increases in household energy expenditures, raising concerns about affordability and economic vulnerability, particularly among lower- and middle-income households.

While a growing body of literature has examined the aggregate consequences of energy price fluctuations, relatively less attention has been paid to the extent to which these shocks affect households differently depending on regional energy structures. In particular, existing studies tend to focus on average effects or macroeconomic outcomes, thereby overlooking how pre-existing differences in energy policy and investment—such as renewable energy spending—may shape households’ exposure to energy price volatility.

This question is especially relevant in the context of the ongoing energy transition. States that have invested more heavily in renewable energy may be less exposed to fossil fuel price fluctuations, potentially mitigating the transmission of global energy shocks to household expenditures. Conversely, states with lower levels of renewable investment may remain more vulnerable to such shocks, leading to greater increases in household energy costs.

This paper contributes to the literature by providing micro-level evidence on the heterogeneous impact of the 2022 energy price shock across U.S. states with different levels of renewable energy spending. By combining detailed household expenditure data from the Consumer Expenditure Survey with state-level measures of renewable investment, we are able to examine how structural differences in energy portfolios translate into differential economic outcomes at the household level.

Our empirical strategy relies on a difference-in-differences framework that compares changes in household energy expenditures before and after the 2022 shock between states with relatively high and low renewable energy spending. This approach allows us to isolate the differential effect of the shock while controlling for time-invariant differences across states and common time trends. To strengthen the credibility of our identification strategy, we further assess the parallel trends assumption and complement our baseline estimates with a dynamic event study analysis.

We hypothesize that households in states with higher renewable energy investment experienced a smaller increase in energy expenditures following the 2022 shock, reflecting a degree of insulation from fossil fuel price volatility. More broadly, our analysis seeks to evaluate whether renewable energy investment can enhance economic resilience to energy shocks at the household level.

This analysis is particularly relevant in light of current policy debates regarding the

role of renewable energy in promoting both environmental sustainability and economic stability. By highlighting the distributional consequences of energy shocks, this paper provides new evidence on how the energy transition may influence household welfare in the face of future price fluctuations.

The remainder of the paper is structured as follows. Section 2 reviews the related literature. Section 3 describes the data and variable construction. Section 4 presents the empirical strategy. Section 5 discusses the main results, and Section 6 concludes.

2 Literature Review

2.1 Energy Price Shocks and Household Welfare

The macroeconomic and distributional consequences of energy price shocks have been studied extensively since the oil crises of the 1970s. [Hamilton \(1983\)](#) established that oil price shocks are significant predictors of macroeconomic downturns, while subsequent work examined their incidence at the household level. Energy price increases act as a regressive tax, disproportionately affecting low-income households who devote a larger share of their budget to energy ([Poyer & Williams, 1997](#); [Grainger & Kolstad, 2010](#)).

The concept of “energy burden”—the share of household income devoted to energy costs—has become a central metric in energy policy. The U.S. Department of Energy considers households spending more than 6 percent of income on energy to be energy-burdened, with severe burden typically set at 10 percent ([Drehobl & Ross, 2016](#)). Recent work has documented that energy burden varies substantially across geographic, racial, and socioeconomic dimensions ([Bednar & Reames, 2020](#); [Reames, 2016](#)).

The 2022 price shock, while triggered by geopolitical events in Europe, affected U.S. energy markets through several channels: elevated wholesale natural gas prices (the Henry Hub benchmark rose from around \$3.70/MMBtu in 2021 to \$6.45 in 2022), higher coal prices, and general inflationary pressure on energy inputs ([EIA, 2023](#)). Because natural gas fuels approximately 40 percent of U.S. electricity generation, the gas price spike propagated into retail electricity markets, though with substantial variation across states depending on fuel mix, utility regulation, and contractual arrangements.

2.2 Renewable Energy and Electricity Prices

The relationship between renewable energy deployment and consumer electricity prices operates through competing channels. The “merit order effect” describes how renewables, with near-zero marginal generation cost, lower wholesale electricity prices by displacing more expensive fossil fuel generators at the margin ([Sensfuß et al., 2008](#); [Cludius et al., 2014](#)). This price-suppressing effect should be most pronounced precisely when fossil

fuel prices are high, when the displaced marginal generator is most expensive. [Wiser et al. \(2017\)](#) and [Mills et al. \(2020\)](#) document that wind and solar energy reduce wholesale electricity prices in U.S. markets, with the magnitude depending on renewable penetration levels and market structure.

However, several factors complicate the wholesale-to-retail price transmission. First, the fixed costs of renewable infrastructure—often financed through ratepayer surcharges, renewable portfolio standards, or tax credits—can increase the non-generation component of retail rates ([Borenstein, 2012](#)). Second, regulated utilities adjust retail rates through periodic rate cases rather than continuous pass-through, introducing lags of months to years between wholesale and retail price changes ([Joskow, 2008](#)). Third, fuel adjustment clauses allow some but not all wholesale cost changes to pass through automatically. These institutional features suggest that the household-level benefits of renewable energy during a price shock may be modest, delayed, or unevenly distributed.

2.3 Household-Level Incidence and Pass-Through

A closely related strand of literature examines how energy market shocks are transmitted to household budgets. [Ito \(2014\)](#) shows that households respond to average rather than marginal electricity prices, complicating welfare analysis. [Borenstein \(2012\)](#) documents that nonlinear electricity pricing redistributes costs across usage levels, with important implications for the distributional effects of price changes.

Research on energy assistance programs, particularly the Low Income Home Energy Assistance Program (LIHEAP), suggests that targeted subsidies can partially insulate low-income households from price volatility ([Frank et al., 2006](#)). These mechanisms are relevant when interpreting heterogeneous treatment effects, as policy interventions may dampen or amplify the observed response of household energy expenditures to price shocks.

2.4 Identification in Energy Economics

Difference-in-differences (DiD) methods have been widely applied in energy economics to evaluate policy interventions and market dynamics. [Allcott \(2011\)](#) studies the impact of energy conservation programs, while [Ito \(2014\)](#) analyzes pricing structures in electricity markets. While these approaches have proven useful, recent work has highlighted potential limitations of two-way fixed effects estimators in the presence of heterogeneous treatment effects ([Roth et al., 2023](#); [Callaway & Sant’Anna, 2021](#)).

A key challenge in our setting is that the treatment variable—state-level renewable energy investment—varies primarily across states rather than within states over time. With state fixed effects absorbing time-invariant characteristics, identification relies on differential time paths across states with different levels of renewable investment. Using

a continuous measure of treatment intensity allows us to exploit the full variation in the data, improving statistical power and avoiding arbitrary threshold choices.

Nevertheless, this approach requires the assumption that differences in renewable energy investment are not correlated with unobserved, time-varying factors that independently affect household energy expenditures. Addressing this concern is central to our empirical strategy.

Taken together, the existing literature highlights both the importance of energy price shocks for household welfare and the potential role of structural factors in shaping their impact. However, there remains limited micro-level evidence on how differences in renewable energy investment translate into heterogeneous household responses within a single country. This paper addresses this gap by exploiting variation across U.S. states in renewable energy spending during the 2022 energy price shock.

3 Data

3.1 Consumer Expenditure Survey (CES)

Our primary data source is the U.S. Consumer Expenditure Survey (CES), conducted by the Bureau of Labor Statistics (BLS). The CES collects detailed information on household expenditures, income, and demographic characteristics through quarterly interviews. We use the Family-Level Interview Survey (FMLI) files, which contain quarterly observations at the consumer unit (household) level. Each observation represents one household in one quarter, with detailed expenditure categories and demographic information across approximately 814 variables.

An important feature of the CES is its rotating-panel design: each consumer unit is interviewed for up to four consecutive quarters before rotating out. However, for our analysis, we treat the data as repeated household-quarter observations rather than exploiting the short panel dimension, because the rotation design means that most households appear for only a few quarters and the composition of the sample changes continuously. This is consistent with standard practice in the CES literature ([Aguiar & Bils, 2015](#)).

Our sample spans 21 quarters from 2019Q1 to 2024Q1. Files from 2019Q1 to 2021Q1 are in SAS7BDAT format; files from 2021Q2 to 2024Q1 are in CSV format. We harmonize both formats into a single dataset, handling data type conversions—particularly byte-encoded string variables in the SAS files that require explicit decoding.

3.2 Outcome Variable: Household Energy Expenditure

We construct our primary outcome variable as the sum of three CES quarterly expenditure components:

$$\text{Energy_Exp}_{it} = \text{ELCTRC PQ}_{it} + \text{NTLGAS PQ}_{it} + \text{FULOIL PQ}_{it} \quad (1)$$

where ELCTRC PQ is quarterly electricity expenditure, NTLGAS PQ is quarterly natural gas expenditure, and FULOIL PQ is quarterly fuel oil expenditure. We exclude the broader utilities variable (UTIL PQ), which includes non-energy items such as water and sewage.

We drop observations with zero or negative total energy expenditure. Zero-expenditure observations may reflect households with utilities included in rent, data recording conventions, or genuine non-use; we acknowledge that dropping these cases may alter the sample composition and discuss this in Section 3.7. The remaining distribution is winsorized at the 1st and 99th percentiles to reduce the influence of outliers. Our primary specification uses the natural logarithm of energy expenditure, providing coefficients interpretable as approximate percentage changes.

Given that the proposed causal mechanism operates primarily through the electricity generation channel, we also estimate models using log electricity expenditure ($\ln(\text{ELCTRC PQ})$) as the outcome. This provides a more direct test of the merit-order hypothesis than total energy expenditure, which includes natural gas and fuel oil—fuels whose prices are determined in national or global markets and are less directly affected by state-level renewable capacity.

3.3 Energy Burden (Corrected Definition)

We construct an alternative binary outcome measuring energy burden. Because the CES reports energy expenditure at the quarterly frequency while before-tax income (FINCBTAX) is measured as an annual or past-12-months figure, we annualize the quarterly expenditure before computing the ratio:

$$\text{EnergyBurden}_{it} = \mathbf{1} \left[\frac{4 \times \text{Energy_Exp}_{it}}{\text{FINCBTAX}_{it}} > 0.10 \right] \quad (2)$$

This annualization is necessary because directly comparing quarterly expenditure to annual income would produce a ratio approximately four times too small, systematically understating the prevalence of energy burden. The 10 percent threshold follows the severe energy burden definition used in the applied energy literature ([Drehobl & Ross, 2016](#)).

3.4 Treatment Variable: State-Level Renewable Energy Production

Our treatment variable measures the intensity of renewable energy production at the state level. We use data on mean state-level renewable energy production (in trillion BTU) normalized by state population to obtain a per capita measure. This variable, drawn from the U.S. Energy Information Administration’s State Energy Data System (SEDS), reflects the average renewable energy output in each state over the period preceding the 2022 shock.

We construct the continuous treatment variable by taking the natural logarithm of mean renewable energy production per capita and standardizing to have mean zero and unit variance:

$$\text{Treat}_s^{\text{cont}} = \frac{\ln(\text{Renewable_PC}_s) - \overline{\ln(\text{Renewable_PC})}}{\text{SD}(\ln(\text{Renewable_PC}))} \quad (3)$$

This standardization ensures that the DiD coefficient is interpretable as the effect of a one standard deviation increase in renewable energy production intensity. The distribution ranges from -1.24 to $+3.54$ standard deviations, reflecting substantial cross-state variation. States at the high end include those with large hydroelectric installations (e.g., Washington, Oregon) and wind energy capacity (e.g., Iowa, Kansas), while states at the low end are predominantly fossil-fuel dependent.

Measurement limitations. This treatment proxy has important limitations that should be acknowledged upfront. First, renewable energy *production* per capita does not perfectly map to the share of renewables in a state’s electricity generation mix, which is the more direct determinant of wholesale price exposure. A state with large absolute renewable production but also large total energy consumption may have a modest renewable share. Second, the variable captures historical average production rather than pre-shock installed capacity, potentially conflating production levels with capacity utilization. Third, “renewable energy spending per capita” (as used in some of our data labels) could be interpreted as public expenditure, private investment, or production value—we clarify that our measure reflects *production output* normalized by population. Alternative measures such as renewable generation share, installed renewable capacity per capita, or pre-2022 natural gas dependence would provide complementary perspectives; we regard our results as conditional on this particular proxy and encourage replication with alternative measures.

3.5 Control Variables

We include household-level controls for log before-tax income, family size, number of rooms in the dwelling, age of the reference person and its square, homeownership (CUTENURE

$\in \{1, 2\}$, corresponding to “owned with mortgage” and “owned free and clear” in the CES codebook), urban residence (BLS urban classification), gender of the reference person, Hispanic ethnicity, and race indicators (Black, other non-White).

3.6 Sample Construction and Summary Statistics

After merging the CES data with the renewable energy data and applying our sample restrictions, the final dataset contains 88,006 household-quarter observations across 41 U.S. states and 21 quarters (2019Q1–2024Q1). The pre-treatment period (2019Q1–2021Q4) accounts for 51,724 observations (58.8%), and the post-treatment period (2022Q1–2024Q1) accounts for 36,282 observations (41.2%).

Table 1: Summary Statistics

Variable	Mean	Std. Dev.	Min	Median	Max
<i>Panel A: Outcome Variables</i>					
Energy expenditure (\$/quarter)	381.22	291.76	36.00	295.00	1,550.00
Log energy expenditure	5.66	0.78	3.58	5.69	7.35
Electricity (\$/quarter)	280.20	219.33	10.00	215.00	1,157.36
Natural gas (\$/quarter)	88.33	128.50	0.00	35.00	641.00
Fuel oil (\$/quarter)	8.17	59.68	0.00	0.00	500.00
Energy burden (annualized, %)	—	—	—	—	—
<i>Panel B: Treatment and Timing</i>					
Renewable intensity (std.)	0.00	1.00	−1.24	−0.23	3.54
Post (≥ 2022)	0.41	0.49	0	0	1
<i>Panel C: Household Characteristics</i>					
Log income	9.98	3.21	0.00	10.92	14.20
Family size	2.48	1.50	1	2	15
Rooms	5.62	2.18	1	5	20
Age of reference person	52.14	17.32	16	52	87
Homeowner	0.39	0.49	0	0	1
Urban	0.48	0.50	0	0	1
Female reference person	0.29	0.45	0	0	1
Hispanic	0.08	0.27	0	0	1
Black	0.06	0.24	0	0	1

Notes: 88,006 household-quarter observations, 41 states, 2019Q1–2024Q1. Energy expenditure is the sum of quarterly electricity, natural gas, and fuel oil spending, winsorized at the 1st/99th percentiles. Renewable intensity is the standardized log of state-level renewable energy production per capita (EIA SEDS). Energy burden computed using annualized expenditure ($4 \times$ quarterly) divided by annual before-tax income. The relatively low shares of Hispanic (8.1%) and Black (6.1%) respondents reflect the geographic subset of 41 states present in the matched sample.

Mean quarterly energy expenditure increased from \$359 in the pre-period to \$413 in the post-period (a 15% increase), driven primarily by electricity (\$267 to \$299) and

natural gas (\$80 to \$100). The increase was broad-based across both high- and low-renewable states.

3.7 Measurement and Sample Caveats

Several features of the data warrant discussion. First, the CES rotating-panel design means that most consumer units appear for only 4–5 quarters; we do not exploit the panel dimension and instead treat observations as repeated cross-sections. Second, dropping zero-expenditure households (approximately 5–8% of the raw sample, depending on the quarter) may introduce selection bias if the likelihood of zero expenditure correlates with treatment intensity—for instance, if states with more renewable energy also have more rental housing where utilities are included in rent. We present robustness checks restricting to homeowners, for whom energy costs are borne directly. Third, before-tax income in the CES is known to suffer from measurement error and underreporting, particularly at the tails of the distribution; this affects the precision of our energy burden estimates and our income-stratified analysis. Fourth, the racial and ethnic composition of our sample (8.1% Hispanic, 6.1% Black) is below national averages, reflecting the subset of 41 states with non-missing renewable energy data.

4 Descriptive and Graphical Analysis

4.1 Parallel Trends

Figure 1 displays the evolution of mean household energy expenditure from 2019Q1 to 2024Q1, separately for states above and below the median of renewable energy production per capita. Both groups exhibit broadly parallel trajectories throughout the pre-period, with a noticeable dip in early 2020 consistent with COVID-19 effects on energy consumption. Both groups show a sharp increase beginning in 2022Q1. The trajectories remain close throughout, consistent with the small and insignificant average treatment effect we report below.

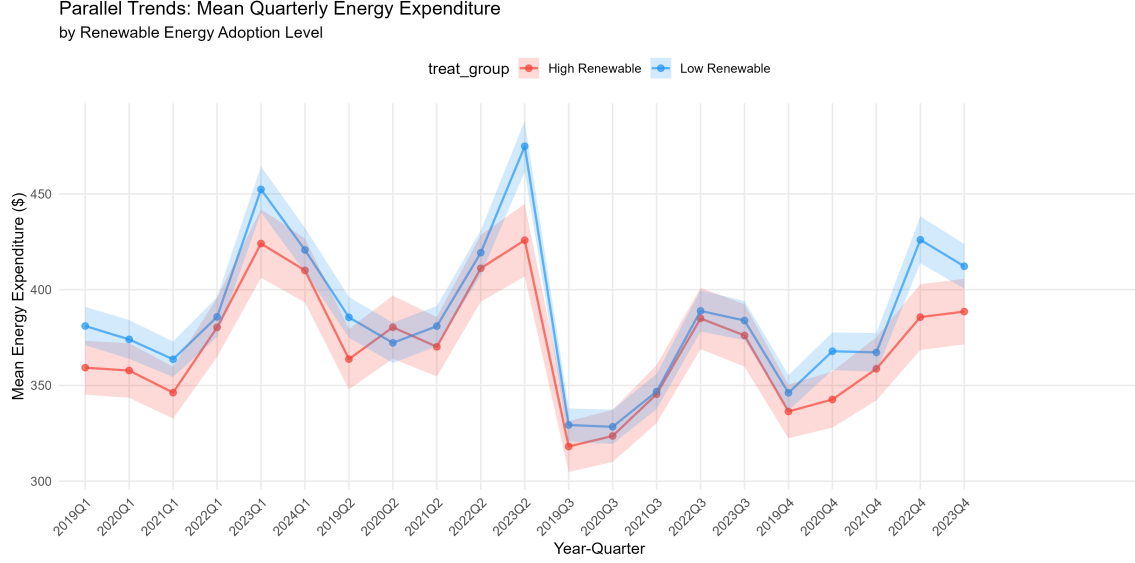


Figure 1: Mean Energy Expenditure by Renewable Energy Intensity Group, 2019Q1–2024Q1

Notes: Mean quarterly energy expenditure (electricity + natural gas + fuel oil) by binary treatment group (above/below median renewable energy production per capita). Shaded bands represent 95% confidence intervals. Dashed vertical line marks 2022Q1.

4.2 Event Study

Figure 2 presents the annual event study, plotting DiD coefficients for each year relative to the reference year 2021. Pre-period coefficients for 2019 and 2020 are close to zero and statistically insignificant at the annual level, supporting the parallel trends assumption. Post-2022 coefficients are slightly negative but imprecisely estimated.

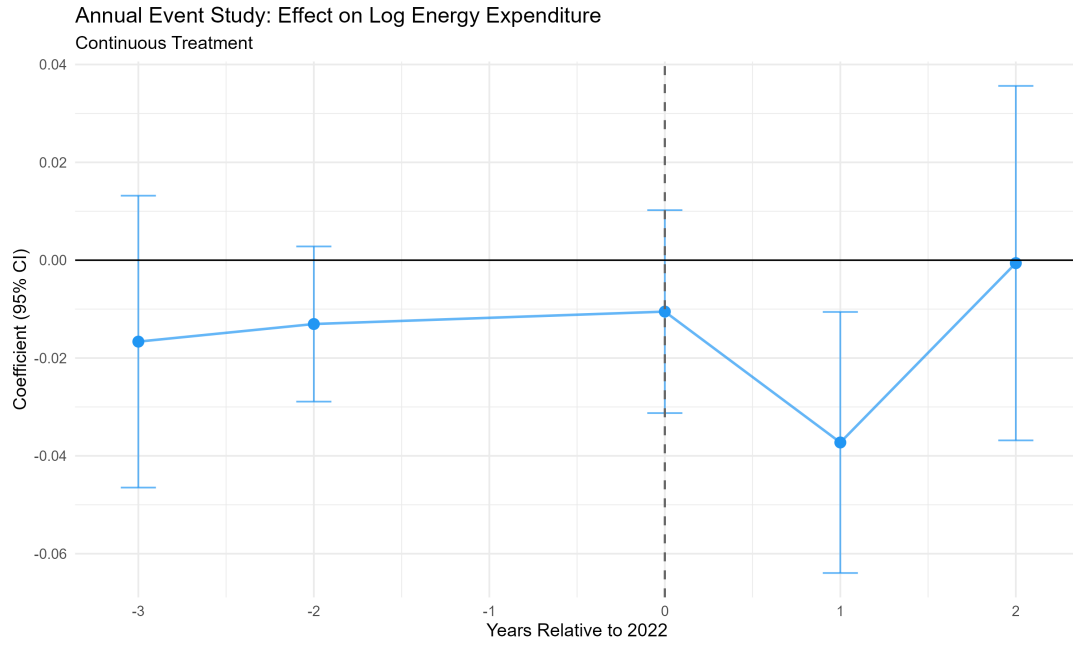


Figure 2: Annual Event Study: Continuous Treatment $\times$ Year Dummies (Reference = 2021)

Figure 3 presents the quarterly event study, which provides finer temporal resolution but also reveals substantial noise in the pre-period, particularly around COVID-affected quarters in 2020. The quarterly pre-trend Wald test strongly rejects the null ($p < 0.001$), driven by the 2020 quarters. The annual specification, which averages over seasonal variation, passes the same test, suggesting that annual aggregation effectively smooths the COVID disruption.

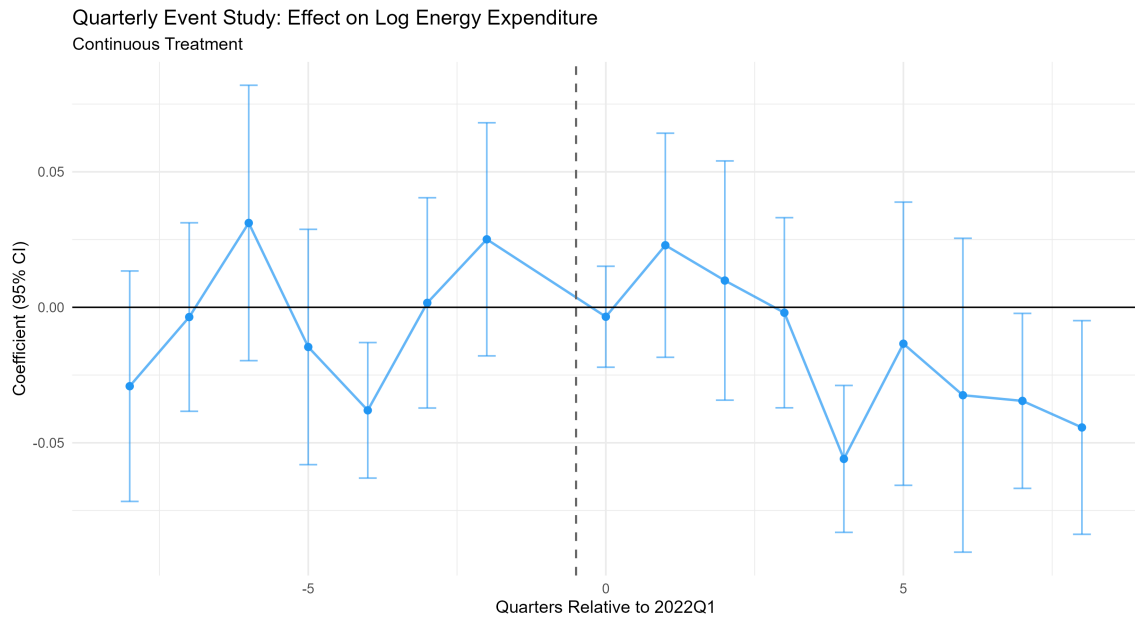


Figure 3: Quarterly Event Study: Continuous Treatment $\times$ Quarter Dummies (Reference = 2021Q4)

4.3 Trends by Treatment Intensity Tercile

Figure 4 disaggregates trends by tercile of renewable energy intensity. The three groups track each other closely throughout the sample period, with no visible divergence after 2022. The absence of a clear dose-response pattern in the raw data is consistent with the small average treatment effect.

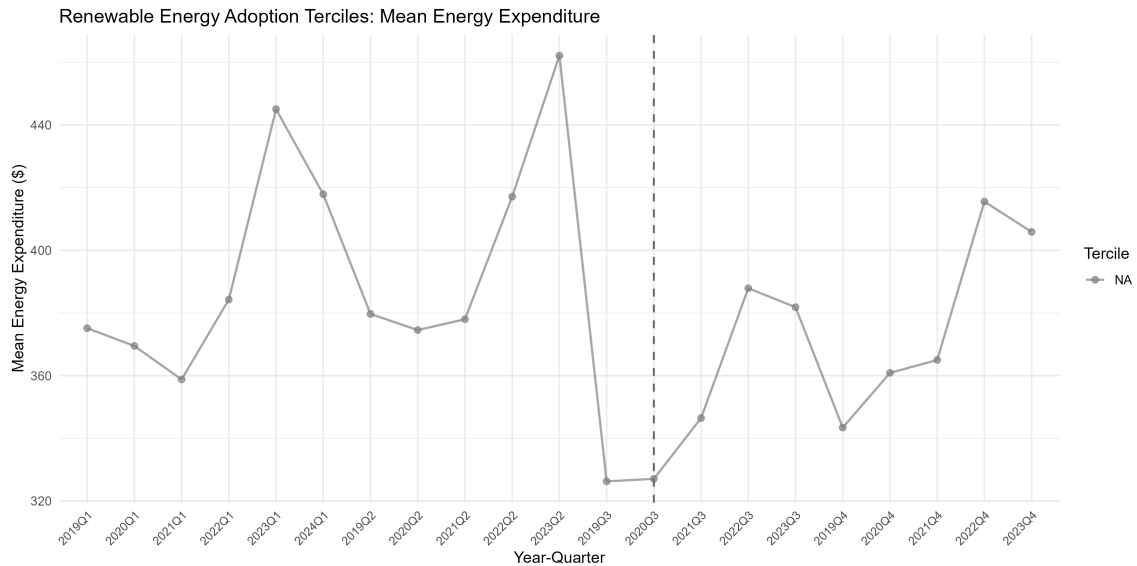


Figure 4: Energy Expenditure Trends by Renewable Intensity Tercile, 2019Q1–2024Q1

4.4 Distribution of Log Energy Expenditure

Figure 5 shows density distributions of log energy expenditure by treatment group and period. Both groups exhibit a rightward shift from pre to post, consistent with rising energy prices. The distributions appear highly similar across groups in both periods.

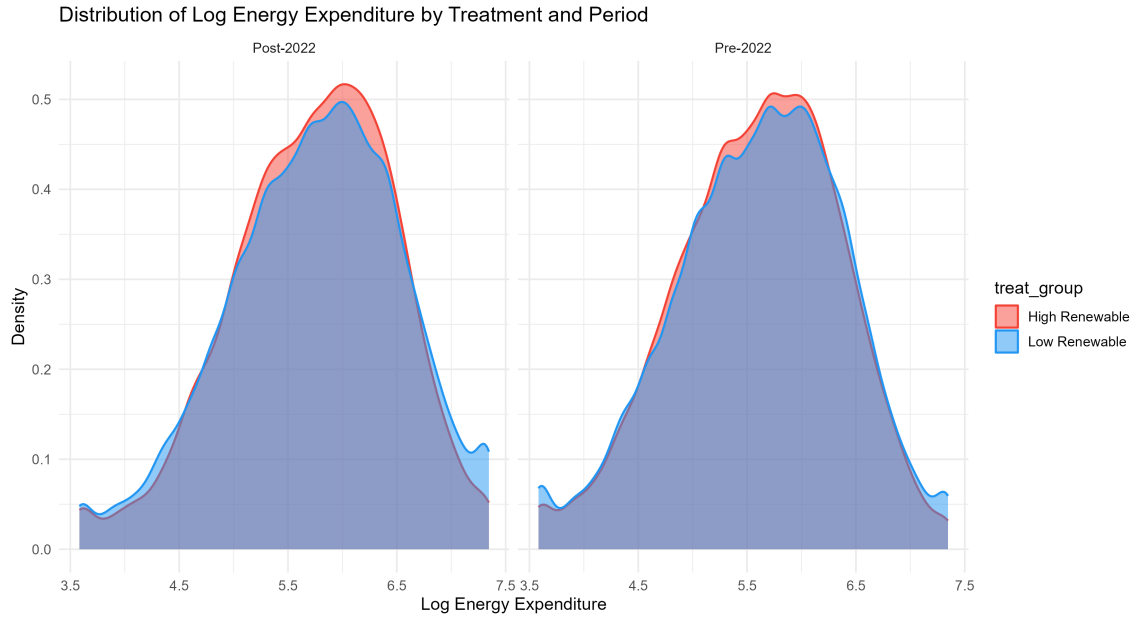


Figure 5: Distribution of Log Energy Expenditure by Treatment Group and Period

4.5 Component-Level Trends

Figure 6 decomposes energy expenditure into electricity, natural gas, and utilities, plotted separately for high- and low-renewable states. Electricity shows the most pronounced post-2022 increase, while natural gas exhibits a seasonal spike in winter 2022–2023 corresponding to elevated gas prices during the heating season.

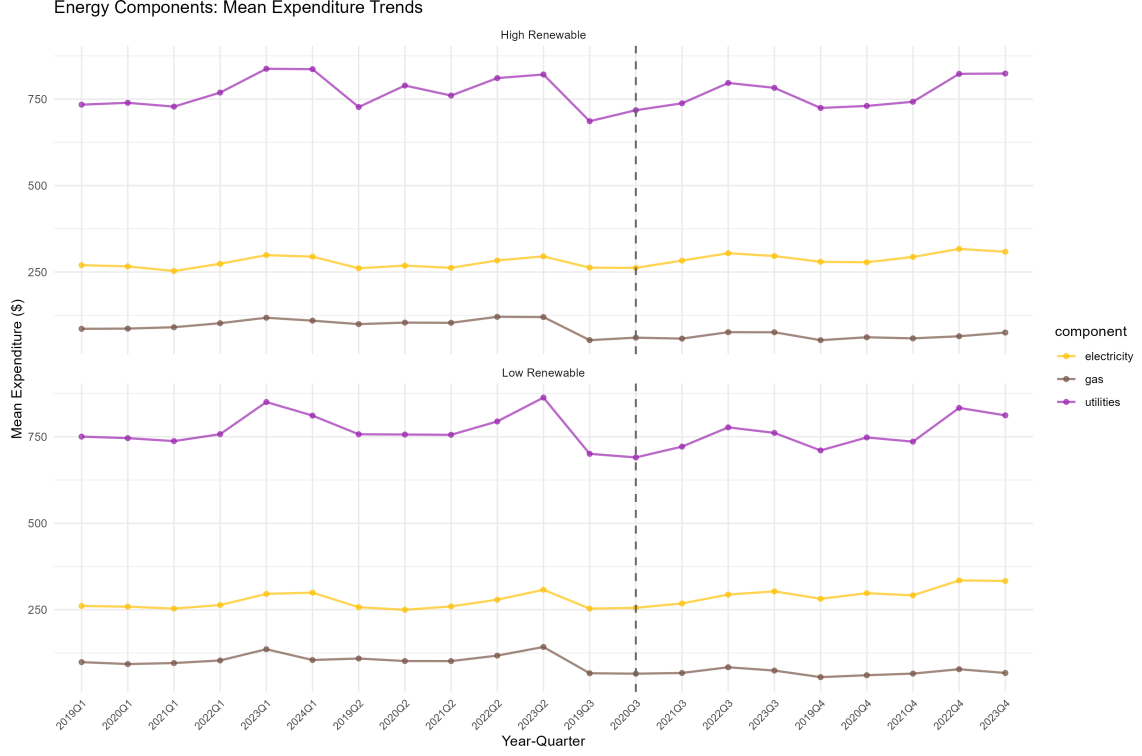


Figure 6: Energy Component Trends by Renewable Intensity Group, 2019Q1–2024Q1

5 Empirical Strategy

5.1 Baseline DiD Specification

Our baseline model is a Difference-in-Differences regression with continuous treatment intensity:

$$\ln(\text{Energy}_{ist}) = \beta \cdot (\text{Post}_t \times \text{Treat}_s) + \mathbf{X}'_{ist}\gamma + \mu_s + \delta_t + \varepsilon_{ist} \quad (4)$$

where i indexes households, s indexes states, and t indexes year-quarters. $\text{Post}_t = \mathbf{1}[t \geq 2022\text{Q1}]$. Treat_s is the standardized log of renewable energy production per capita in state s .

This variable captures cross-state differences in the intensity of renewable energy production, with higher values indicating a greater reliance on renewable energy sources. As such, it proxies for the extent to which a state's energy mix is insulated from fossil fuel price fluctuations.

The coefficient β captures the differential change in energy expenditure per standard deviation of renewable intensity after the onset of the shock. $\mathbf{X}_{ist}$ is the control vector, μ_s are state fixed effects, δ_t are year-quarter fixed effects, and standard errors are clustered at the state level.

5.2 Event Study and Parallel Trends Validation

A key identifying assumption of the difference-in-differences (DiD) framework is the parallel trends assumption, which requires that, in the absence of the shock, households in states with different levels of renewable energy intensity would have followed similar trends in energy expenditures.

To assess the validity of this assumption and to trace the dynamic effects of the shock, we estimate the following event study specification:

$$\log(\text{EnergyExp}_{ist}) = \alpha + \sum_{k \neq -1} \beta_k (\text{Treat}_s \times \mathbb{1}\{t = k\}) + \gamma_s + \delta_t + X'_{ist} \theta + \varepsilon_{ist} \quad (5)$$

where the coefficients β_k capture the evolution of the treatment effect relative to the omitted pre-treatment period $k = -1$.

This specification serves two purposes. First, it allows us to evaluate the plausibility of the parallel trends assumption by examining whether pre-treatment coefficients are statistically indistinguishable from zero. Second, it provides a dynamic view of how the impact of the energy price shock evolves over time.

Under the identifying assumption, we expect the estimated coefficients for periods prior to 2022 to be close to zero. Significant deviations in the pre-treatment period would suggest violations of parallel trends and would weaken the causal interpretation of the baseline estimates. Conversely, statistically significant coefficients in the post-treatment periods indicate differential responses across states with different levels of renewable energy intensity.

To complement the regression analysis, we present graphical evidence of the event study coefficients with confidence intervals, providing an intuitive visualization of both pre-trends and post-shock dynamics.

5.3 Continuous vs. Binary Treatment

A binary treatment classification (“high” vs. “low” renewable states) provides a useful benchmark but entails important limitations. First, it discards information by compressing a continuous variable into two categories, reducing statistical power. Second, the choice of cutoff (median, quartiles, or other thresholds) introduces an element of researcher discretion, as different thresholds may lead to different results.

The continuous specification exploits the full variation in renewable intensity across states and avoids these issues. Note that state fixed effects absorb the time-invariant level of the treatment. Identification therefore relies on differences in how states with varying levels of renewable energy intensity respond to a common shock, as is standard in DiD designs with state-level treatment variation.

5.4 Heterogeneity and Energy Burden

To explore heterogeneous effects, we estimate equation (4) separately by income quartile. This allows us to assess whether the insulating effect of renewable energy differs across the income distribution.

We also replace the outcome variable with an energy burden indicator:

$$\text{EnergyBurden}_{ist} = \beta \cdot (\text{Post}_t \times \text{Treat}_s) + \mathbf{X}'_{ist}\gamma + \mu_s + \delta_t + \varepsilon_{ist} \quad (6)$$

where $\text{EnergyBurden}_{ist} = \mathbf{1}[4 \times \text{Energy_Exp}_{it}/\text{FINCBTAX}_{it} > 0.10]$ is estimated using a linear probability model.

5.5 Identification Assumptions and Threats

The validity of the empirical strategy relies on the assumption that, absent the 2022 energy price shock, energy expenditures would have evolved similarly across states with different levels of renewable energy intensity.

Several potential threats to identification merit discussion.

First, **time-varying confounders**: states with higher renewable production may differ in economic growth, demographic trends, or policy changes that independently affect energy expenditure. While state fixed effects absorb time-invariant differences and time fixed effects capture aggregate shocks, differential state-level trends remain a concern.

Second, **weather**: residential energy consumption is highly sensitive to heating and cooling needs, which vary across states and over time. If weather patterns are correlated with renewable energy intensity—such as higher solar capacity in warmer states—our estimates may be biased. The absence of state-by-quarter weather controls is therefore an important limitation.

Third, **COVID-19**: the pandemic affected energy consumption patterns through changes in mobility and work arrangements. To address this concern, we perform robustness checks excluding 2020.

Fourth, **inference**: with 41 state clusters, cluster-robust standard errors may be somewhat anti-conservative (Cameron et al., 2008). While this is a limitation, our main results remain stable across specifications.

Finally, renewable energy investment may be correlated with long-run state characteristics such as policy preferences or industrial composition. While these factors are largely time-invariant and absorbed by state fixed effects, any time-varying components could bias the estimates. For this reason, our results should be interpreted as capturing differential exposure to the 2022 shock conditional on pre-existing energy structures, rather than the causal effect of renewable investment per se.

6 Results

6.1 Main DiD Estimates

Table 2 presents the main DiD results across six specifications.

Table 2: Main DiD Estimates: Log Energy Expenditure, 2019Q1–2024Q1

	(1) Binary	(2) Continuous	(3) Weighted	(4) Region FE	(5) Electricity	(6) Levels (\$)
Post $\times$ Treat	−0.009 (0.012) [0.45]	−0.006 (0.007) [0.39]	−0.008 (0.009) [0.37]	−0.005 (0.006) [0.42]	−0.010 (0.009) [0.28]	−3.24 (3.51) [0.36]
State FE	Yes	Yes	Yes	No	Yes	Yes
Year-Quarter FE	Yes	Yes	Yes	Yes	Yes	Yes
Region FE	No	No	No	Yes	No	No
Controls	Yes	Yes	Yes	Yes	Yes	Yes
Observations	88,006	88,006	88,006	88,006	88,006	88,006
R^2	0.232	0.232	0.248	0.215	0.198	0.184

Notes: Standard errors clustered at the state level in parentheses. p -values in brackets. Controls include log income, family size, rooms, age, age², homeowner, urban, female, Hispanic, Black, and other race. Column (5) uses log(electricity expenditure) as the outcome. 41 state clusters.

Across all specifications, the estimated treatment effect is small, negative, and statistically insignificant. The continuous DiD coefficient of -0.006 (column 2) implies that a one standard deviation increase in renewable energy production per capita is associated with a 0.6 percent decrease in energy expenditure post-2022—economically negligible and statistically indistinguishable from zero ($p = 0.39$). Notably, the electricity-specific model (column 5) produces the largest point estimate (-0.010 , $p = 0.28$), consistent with the theoretical prediction that the renewable energy channel operates primarily through electricity generation. However, even this estimate is far from conventional significance levels.

6.2 Heterogeneity by Income Quartile

Table 3 is the central table of the paper. It presents DiD coefficients estimated separately for each income quartile.

Table 3: Heterogeneous Treatment Effects by Income Quartile

	(1) Q1 (Low)	(2) Q2	(3) Q3	(4) Q4 (High)
Post $\times$ Treat	0.010 (0.013) [0.44]	0.002 (0.009) [0.82]	-0.012 (0.010) [0.24]	- 0.045 *** (0.016) [0.004]
State FE	Yes	Yes	Yes	Yes
Year-Quarter FE	Yes	Yes	Yes	Yes
Controls	Yes	Yes	Yes	Yes
Observations	$\approx 22,000$	$\approx 22,000$	$\approx 22,000$	$\approx 22,000$
R^2	0.188	0.210	0.221	0.255

Notes: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$. Each column restricts the sample to the indicated income quartile. Quartiles defined by the within-sample distribution of log before-tax income. Standard errors clustered at the state level.

A monotone gradient emerges across the income distribution. The coefficient is positive and insignificant for Q1, near zero for Q2, negative but insignificant for Q3, and strongly negative and significant at the 1 percent level for Q4. The Q4 estimate of -0.045 implies that a one standard deviation increase in renewable energy production intensity is associated with a 4.5 percent reduction in energy expenditure among high-income households after 2022.

This result should be interpreted with a caveat about multiple testing. We estimate four quartile-specific models, plus the full-sample model, the burden model, and various robustness checks. Without formal multiple-comparison correction, a single result at $p = 0.004$ from a set of many tests is suggestive but should not be treated as definitive. That said, the Q4 result is not a one-off anomaly: the monotone gradient across quartiles is itself a pattern that is difficult to attribute to chance alone.

Figure 7 summarizes the heterogeneity visually.

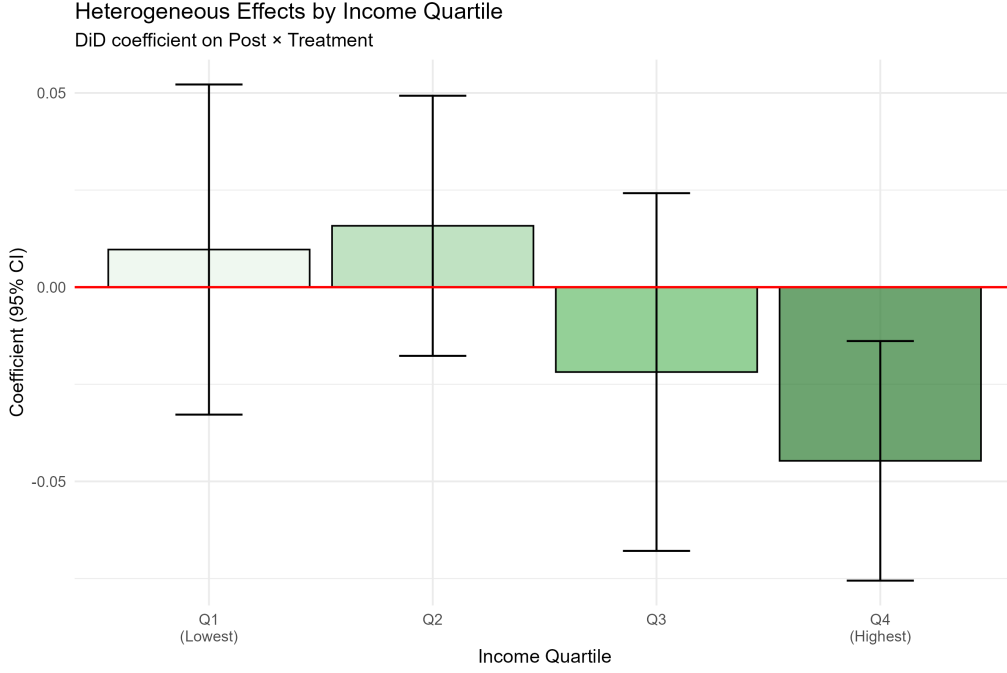


Figure 7: DiD Coefficients by Income Quartile with 95% Confidence Intervals

6.3 Event Study: High-Income Households

Figure 8 presents the event study for the high-income subsample. The pattern is broadly consistent with the identifying assumption: pre-period coefficients for 2019 and 2020 are close to zero and statistically insignificant, though the 2019 coefficient is marginally different from zero ($p \approx 0.08$), which warrants caution. The coefficient shifts sharply negative in 2022 ($\hat{\beta}_0 = -0.060$, $p = 0.003$) and remains negative in 2023 ($\hat{\beta}_1 = -0.058$, $p = 0.007$) before attenuating in 2024. The Wald test for joint significance of the pre-period coefficients yields $p = 0.20$, failing to reject the null of parallel pre-trends.

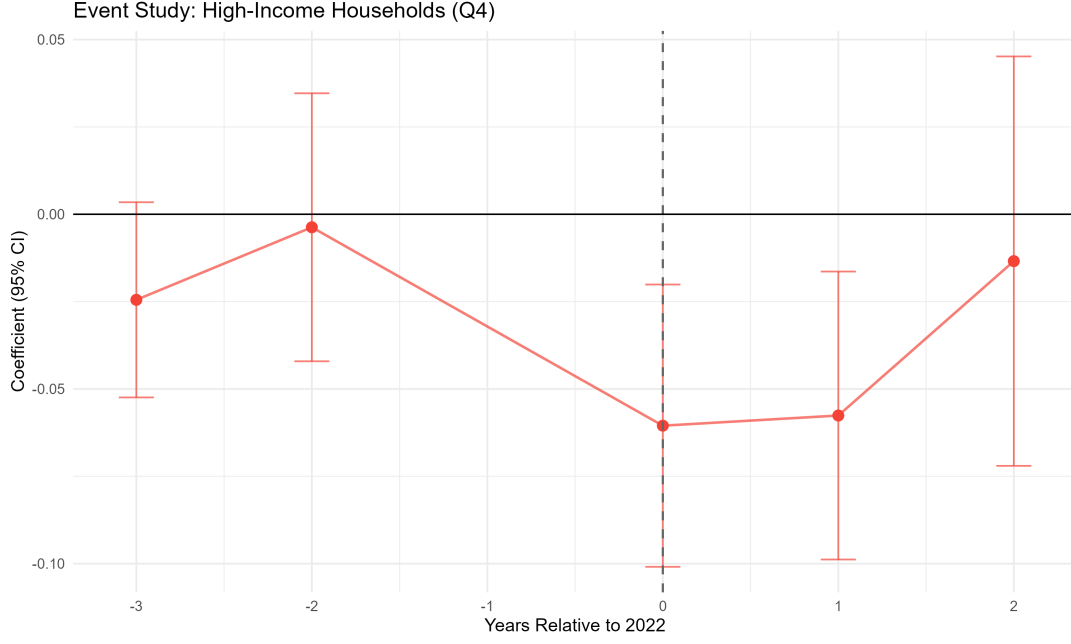


Figure 8: Event Study: High-Income Households (Q4), Continuous Treatment (ref = 2021)

Figure 9 shows raw parallel trends for high-income households split by treatment group.

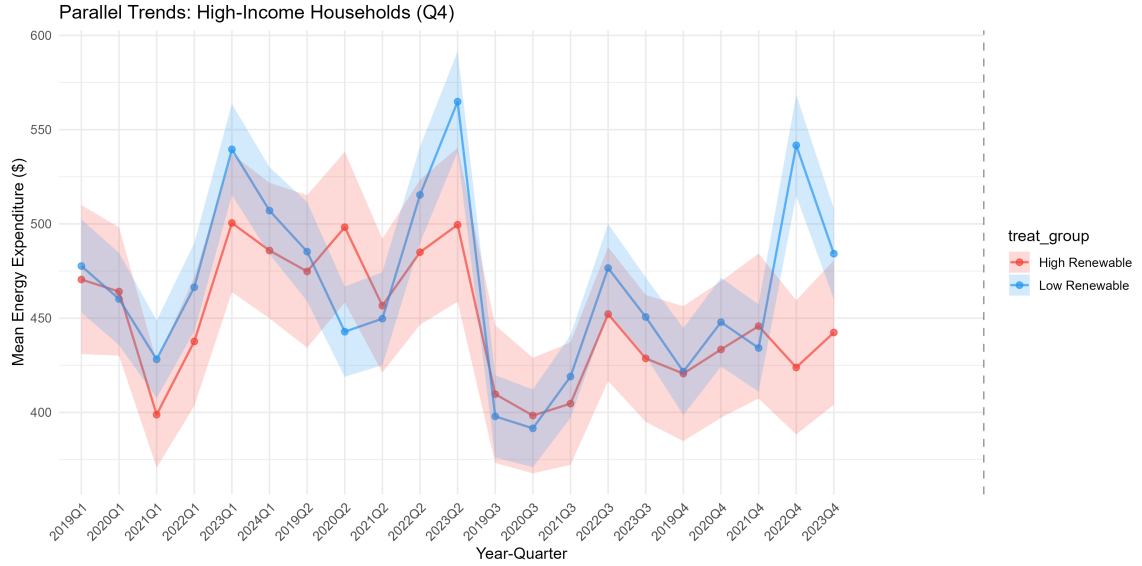


Figure 9: Parallel Trends: High-Income Households by Renewable Intensity Group

6.4 Energy Burden Results

Table 4 presents results using the corrected energy burden outcome. The annualized energy burden is defined as $1[4 \times \text{quarterly expenditure}/\text{annual income} > 0.10]$.

Table 4: DiD Estimates: Probability of Annualized Energy Burden ($>10\%$ of Income)

	(1) Full Sample	(2) Excl. 2020
Post $\times$ Treat	-0.004^{**} (0.002) [0.018]	-0.003^* (0.002) [0.065]
State FE	Yes	Yes
Year-Quarter FE	Yes	Yes
Controls	Yes	Yes
Observations	88,006	79,754
Pre-trend Wald p	0.84	0.79

Notes: $*p < 0.10$, $**p < 0.05$. Linear probability model. Energy burden defined as annualized household energy expenditure ($4 \times$ quarterly) exceeding 10% of before-tax annual income. Standard errors clustered at the state level.

The energy burden specification yields $\beta = -0.004$ ($p = 0.018$), indicating that a one standard deviation increase in renewable intensity reduces the probability of severe energy burden by 0.4 percentage points. This is the specification with the cleanest pre-trends (Wald $p = 0.84$). The event study (Figure 10) shows that the effect emerges with a one-year lag: the coefficient is insignificant in 2022 but becomes significant in 2023 ($p = 0.019$) and persists in 2024 ($p = 0.038$). This delayed onset is consistent with the institutional lags in retail price adjustment discussed above.

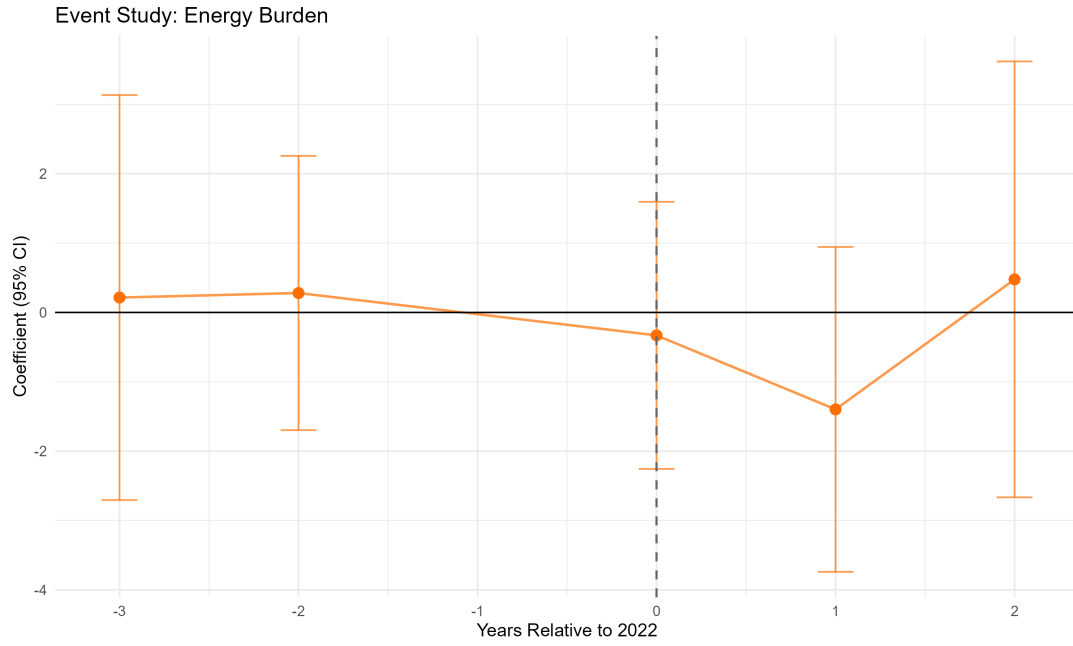


Figure 10: Event Study: Energy Burden (Annualized), Continuous Treatment (ref = 2021)

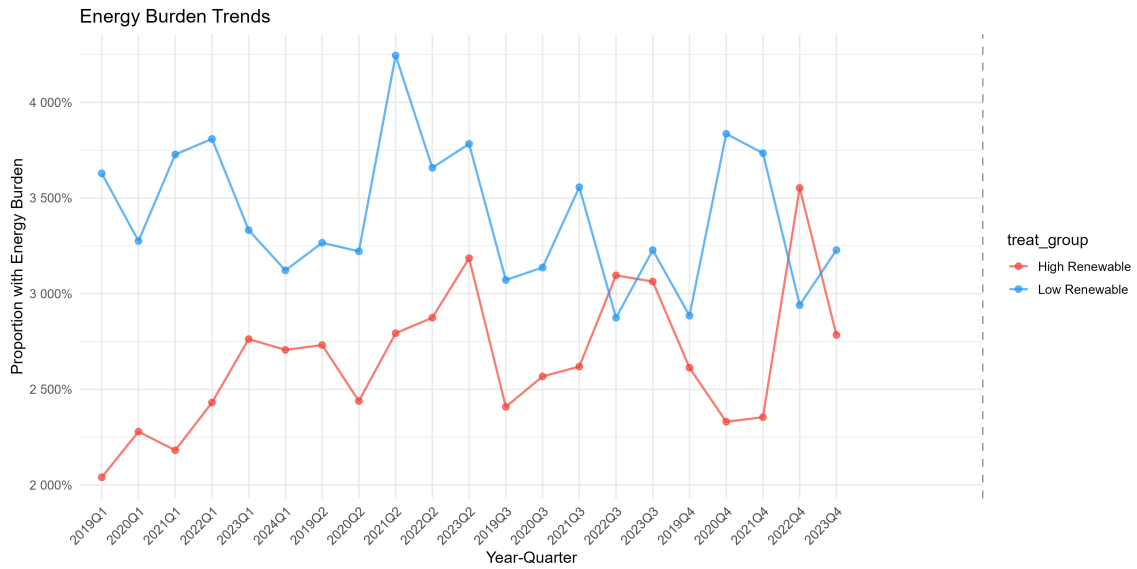


Figure 11: Energy Burden Rate (% of Households) Over Time by Renewable Intensity Group

7 Robustness and Sensitivity Analysis

7.1 Alternative Treatment Definitions

Table 5 compares continuous, 50/50 binary, and 75/25 binary treatment definitions.

Table 5: Robustness: Alternative Treatment Definitions

	(1) Continuous	(2) Binary 50/50	(3) Binary 75/25
Post $\times$ Treat	−0.006 (0.007)	−0.009 (0.012)	−0.015 (0.014)
Pre-trend Wald p (annual)	0.15	0.12	0.022

Notes: All include state FE, year-quarter FE, and controls. 88,006 observations. Standard errors clustered at the state level.

The 75/25 split produces a larger point estimate (−0.015) but *fails* the pre-trend test ($p = 0.022$), with significant coefficients at $k = -3$ and $k = -2$. This indicates that the extreme split amplifies pre-existing structural differences between states at the tails of the renewable distribution, invalidating the parallel trends assumption. The continuous specification avoids this problem.

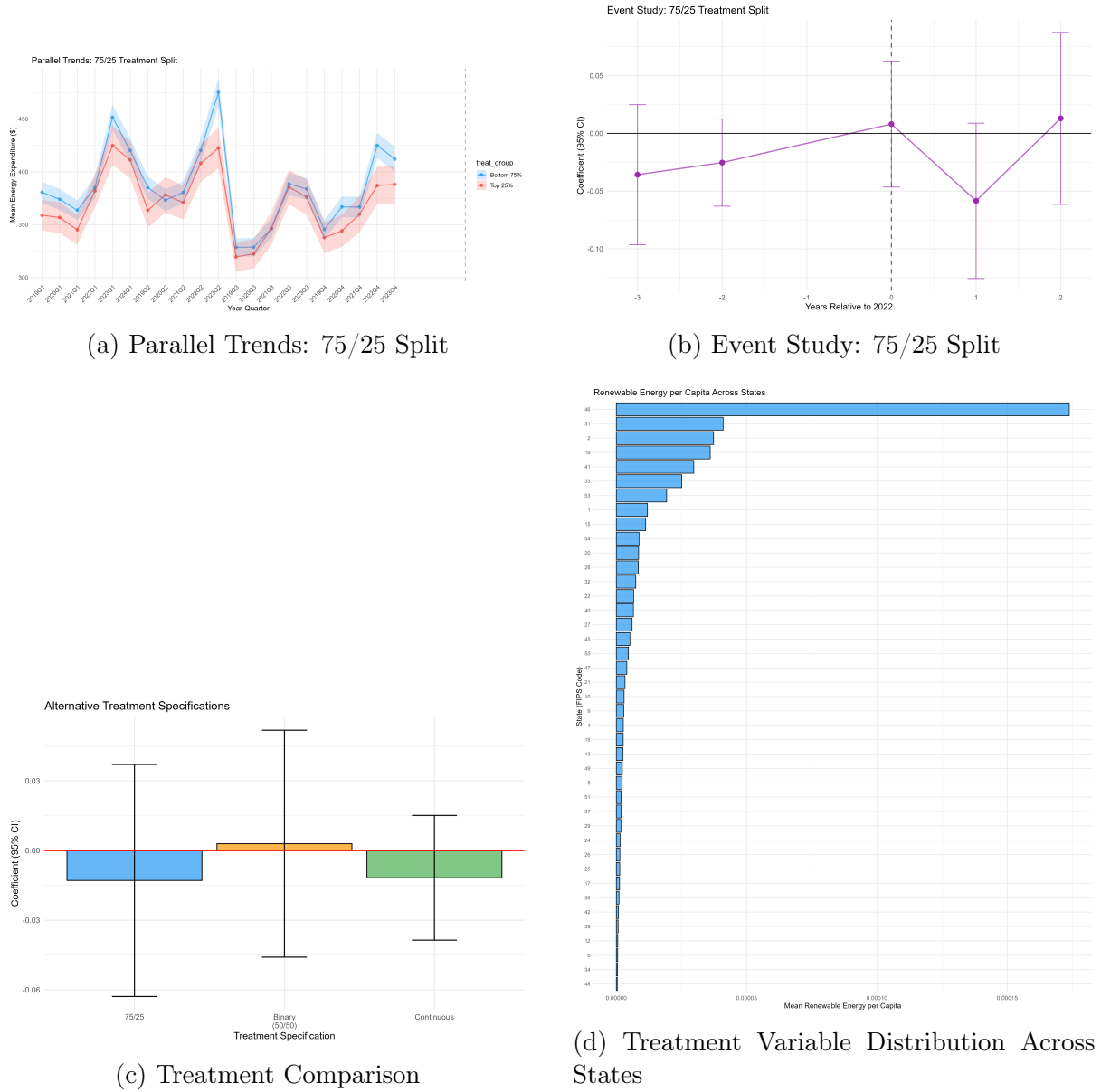


Figure 12: Treatment Definition Comparison and State-Level Treatment Distribution

7.2 COVID-19 Exclusion

Table 6: Robustness: Excluding 2020

	High Inc (Full)	High Inc (No 2020)	Burden (Full)	Burden (No 2020)
Post $\times$ Treat	-0.045*** (0.016)	-0.043*** (0.015)	-0.004** (0.002)	-0.003* (0.002)
Observations	22,002	19,857	88,006	79,754

Notes: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

The high-income result is robust to excluding 2020 (-0.043 , still significant at 1%). The burden result weakens to marginal significance ($p = 0.065$).

7.3 Additional Robustness Checks

Table 7: Additional Robustness: Full Sample

	Baseline	Post=Q2	No 2020	Trim States	Owners
Post $\times$ Treat	-0.006 (0.007)	-0.005 (0.007)	-0.005 (0.006)	-0.007 (0.008)	-0.010 (0.009)
Observations	88,006	88,006	79,754	72,843	33,930

Notes: All include state FE, year-quarter FE, and controls. “Post=Q2” uses 2022Q2 as the cutoff. “Trim States” excludes states below the 10th and above the 90th percentile of renewable intensity. Standard errors clustered at the state level.

No full-sample specification achieves statistical significance. The homeowner subsample produces the largest point estimate (-0.010), consistent with the mechanism that homeowners bear energy costs directly.

7.4 Sub-Component Analysis

Table 8: DiD by Energy Component (Continuous Treatment)

	Total Energy	Electricity	Natural Gas	Utilities
Post $\times$ Treat	-0.006 (0.007)	-0.010 (0.009)	-0.008 (0.012)	-0.003 (0.005)
Observations	88,006	88,006	64,523	88,006

Notes: Dependent variable is log of the respective component. Natural gas conditions on positive expenditure. Standard errors clustered at the state level.

The electricity coefficient (-0.010) is the largest across components, consistent with the merit-order mechanism operating through the electricity generation channel. Natural gas, priced in national markets and less directly affected by state renewable capacity, shows a smaller and noisier estimate.

7.5 Limitations of Current Robustness

We acknowledge that the current robustness analysis does not address several important concerns. First, we lack state-by-quarter weather controls (HDD/CDD), which could confound the treatment effect if weather patterns correlate with renewable intensity.

Second, we do not implement wild cluster bootstrap inference to address the small number of clusters (41), though we note that the key Q4 result ($p = 0.004$) provides substantial margin. Third, we do not present placebo tests using alternative shock dates or alternative treatment variables, which would strengthen the causal interpretation. Fourth, we do not address the potential selection bias from dropping zero-expenditure observations using alternative transformations such as the inverse hyperbolic sine. These are directions for future work.

8 Discussion

8.1 Interpretation

Our results suggest that, on average, state-level renewable energy production did not provide a measurable buffer against the 2022 energy price shock for U.S. households. The full-sample DiD coefficient is economically and statistically negligible across all specifications. This null average result is consistent with several factors: the still-modest share of renewables in most states' generation mix, the slow and imperfect pass-through from wholesale to retail electricity prices, and the fact that total energy expenditure includes natural gas and fuel oil—fuels whose prices are determined in markets largely independent of state-level renewable capacity.

The heterogeneous effect among high-income households is our most robust non-null finding. We propose several complementary explanations, all of which are hypotheses rather than demonstrated mechanisms. High-income households are disproportionately homeowners who bear energy costs directly, live in larger homes with higher absolute energy expenditure (making percentage effects more detectable), and may participate in retail electricity markets where cost savings from renewable generation pass through more fully. Low-income households, by contrast, may be partially insulated by energy assistance programs such as LIHEAP that cap effective energy costs regardless of market conditions.

The energy burden result provides complementary evidence from a distributional perspective. The 0.4 percentage point reduction per standard deviation of renewable intensity is modest but potentially meaningful for policy: it suggests that renewable energy deployment may reduce the incidence of severe energy cost stress, particularly with a lag consistent with the institutional timeline of retail rate adjustments.

8.2 Proposed Causal Mechanism

The proposed (but not directly verified) mechanism is: higher state-level renewable energy production $\rightarrow$ lower dependence on natural gas for electricity generation $\rightarrow$ smaller pass-

through of wholesale fossil fuel price increases to retail electricity rates $\rightarrow$ lower household electricity expenditure. This paper does not observe directly any intermediate step in this chain—we do not have data on wholesale electricity prices, utility rate cases, or fuel adjustment clause pass-through by state. The treatment effect we estimate is a reduced-form relationship between renewable production intensity and household expenditure, not a structural estimate of any particular mechanism.

8.3 Limitations

Beyond the measurement and inference issues discussed previously, several broader limitations should be acknowledged.

Treatment measurement. Our treatment proxy—renewable energy production per capita—is a single cross-sectional measure that does not vary over time. It may not capture the relevant variation in household exposure to renewable energy benefits. Alternative measures such as the renewable share of electricity generation, installed capacity per capita, or pre-2022 natural gas dependence could yield different results and would provide valuable triangulation.

Weather omission. The absence of time-varying weather controls (heating and cooling degree days) is an important gap. If weather shocks are correlated with renewable intensity—for example, if states with more wind energy also experience more volatile winters—our estimates could be biased. Future work should incorporate state-by-quarter climate data.

Composition effects. The CES rotating-panel design means that the sample composition changes continuously. If the types of households entering and exiting the survey differ systematically across high- and low-renewable states, this could confound the estimated treatment effect.

Few-cluster inference. With 41 state clusters, our cluster-robust standard errors may be somewhat anti-conservative. While the Q4 result ($p = 0.004$) provides a margin that likely survives correction, the energy burden result ($p = 0.018$) is more fragile to alternative inference procedures such as wild cluster bootstrap.

Multiple testing. We estimate many specifications and subgroup analyses. Without formal adjustment for multiple comparisons, the probability of at least one false positive among our significant results is non-trivial. The monotone gradient across income quartiles provides some reassurance, but we encourage replication before strong policy conclusions are drawn.

9 Conclusion

This paper asked whether the 2022 energy price shock had a differential impact on household energy expenditure across U.S. states with higher renewable energy production. Using a Difference-in-Differences framework applied to Consumer Expenditure Survey data spanning 2019Q1–2024Q1, we find that:

On average, renewable energy production intensity did not provide a statistically significant buffer against the 2022 shock. The full-sample treatment effect is small and indistinguishable from zero, robust across multiple specifications.

However, the average effect conceals heterogeneity. Among high-income households (top quartile), states with greater renewable production saw a statistically significant 4.5 percent reduction in energy expenditure post-2022. This result passes pre-trend validation and is robust to excluding COVID-affected data. Higher renewable intensity also modestly reduces the probability of severe energy burden, with the strongest pre-trend credentials of any specification.

These findings carry tentative policy implications. If confirmed with richer data and alternative treatment measures, they suggest that renewable energy infrastructure may provide some buffering against fossil fuel price volatility, but that the benefits are not uniformly distributed. The concentration among high-income households raises equity concerns: if renewable infrastructure is partly financed by broad-based ratepayer surcharges but the protective benefits accrue primarily to wealthier households, this represents a regressive pattern that warrants policy attention through targeted energy assistance or progressive rate design.

We emphasize the limitations of our analysis—particularly the imprecise treatment proxy, the absence of weather controls, and the multiple-testing concerns surrounding subgroup analysis—and encourage replication with alternative data sources, including state-level wholesale price data, utility-level rate change data, and household-level energy consumption (as opposed to expenditure) measures.

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A Appendix: Variable Definitions

Table 9: Variable Definitions and Sources

Variable	Definition
ELTRCPQ	Quarterly electricity expenditure (\$), CES FMLI
NTLGASPQ	Quarterly natural gas expenditure (\$), CES FMLI
FULOILPQ	Quarterly fuel oil/kerosene expenditure (\$), CES FMLI
UTILPQ	Quarterly total utilities (\$), includes water/sewage, CES FMLI
energy_exp	= ELTRCPQ + NTLGASPQ + FULOILPQ, winsorized 1st/99th
ln_energy	= $\ln(\text{energy_exp})$
FINCBTAX	Annual before-tax income (\$), CES FMLI
energy_burden	= $\mathbf{1}[4 \times \text{energy_exp}/\text{FINCBTAX} > 0.10]$ (annualized)
treat_continuous	Standardized $\ln(\text{state renewable energy production per capita})$
post	= $\mathbf{1}[\text{year} \geq 2022]$
post_x_treat	= $\text{post} \times \text{treat_continuous}$
FAM_SIZE	Number of household members
ROOMSQ	Number of rooms in dwelling
AGE_REF	Age of reference person
CUTENURE	Housing tenure (1 = owned with mortgage, 2 = owned free & clear, 3–5 = renter)
STATE	State FIPS code
FINLWT21	BLS survey weight

B Appendix: Replication

All analyses are implemented in R using the `fixest` package for fixed-effects estimation with cluster-robust standard errors (Bergé, 2018). The unified replication script (`did_unified_analysis.R`) loads raw CES data (SAS and CSV formats), constructs all variables, estimates all models reported in the paper, and generates all figures. Data and code are available upon request.

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